

Why Do Some Canadian Provinces Have Higher Hospital Mortality Despite Stable National Trends?



Exploring Provincial Variation in Hospital Standardized Mortality Ratio (HSMR)

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THE PROBLEM

- Hospital mortality varies significantly across Canadian provinces.
- National trends appear stable, but regional disparities remain persistent.
- Some provinces consistently exceed expected mortality levels (HSMR >100).
- Understanding these differences is essential for improving outcomes and equity.



KEY TAKEAWAY



Hospital mortality in Canada remains relatively stable overall; however, significant inter-provincial variation persists.

Certain provinces—particularly in Atlantic regions—consistently report **HSMR >100**, highlighting the need for **targeted, data-driven interventions**.



WHAT IS HSMR?

HSMR compares the actual number of in-hospital deaths to the number expected, adjusted for patient characteristics.

HSMR < 100
Lower than expected mortality

HSMR = 100
Expected mortality

HSMR > 100
Higher than expected mortality



WHY DATA MATTERS



Supports targeted policy interventions



Guides resource allocation to high-risk regions



Enables real-time performance monitoring



Improves health system accountability and quality

KEY FINDINGS



Persistent Variation: Significant differences in HSMR across provinces.



High-Risk Regions: NL, PE, and MB consistently report HSMR >100.

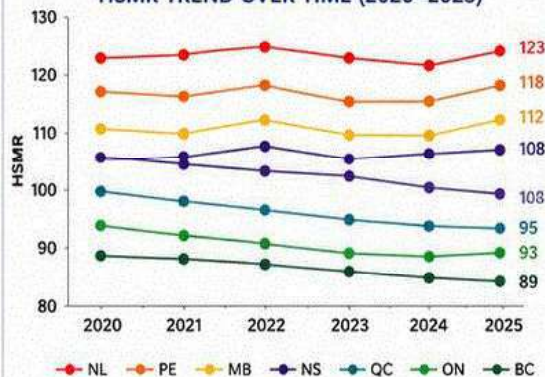


Improvement Trend: Ontario and British Columbia show steady decline and improvement.



Atlantic Disparity: Atlantic provinces remain above national benchmark.

HSMR TREND OVER TIME (2020–2025)



Ontario and British Columbia show consistent improvement over time and are below expected mortality (HSMR <100).

HSMR BY PROVINCE (2025)



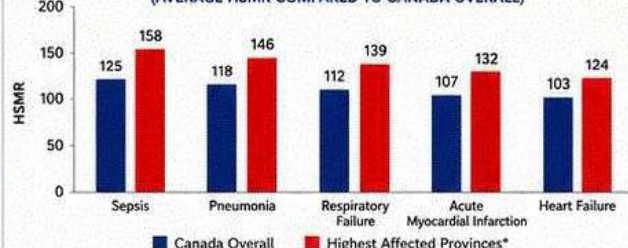
- BC (89) now matches the trend chart.
- Ontario (93) and British Columbia (89) are both below expected mortality, which is why both are green.



CHART INSIGHTS

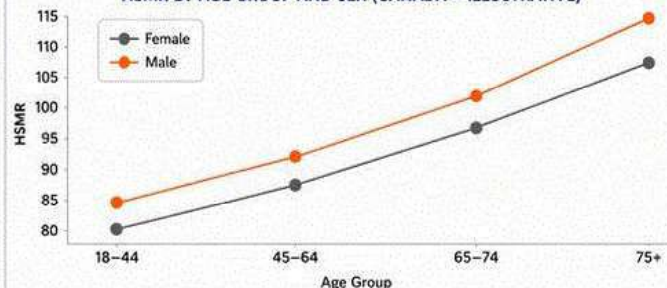
- BC value corrected to 89 in the map to match the trend chart.
- Color coding is consistent: all provinces with HSMR <100 are green.
- The trend chart focuses on key provinces for better clarity.
- Atlantic provinces consistently exceed HSMR >100, indicating higher risk.

TOP DIAGNOSTIC GROUPS CONTRIBUTING TO HIGHER MORTALITY (AVERAGE HSMR COMPARED TO CANADA OVERALL)



*Provinces with highest average HSMR in each diagnostic group (2020–2025).

HSMR BY AGE GROUP AND SEX (CANADA – ILLUSTRATIVE)



INTERPRETATION OF VARIATION

Differences in HSMR may reflect:

- Variations in healthcare access and service availability
- Difference in hospital capacity and system pressure
- Population health status and demographics
- Variability in clinical practices and care quality



DATA LIMITATIONS

HSMR is adjusted for age and sex, but may not fully account for:

- Comorbidities
- Coding variability
- Documentation differences

Results are descriptive and should be interpreted cautiously.



IMPORTANT NOTE

“Findings are descriptive and do not establish causal relationships.”



SYSTEM CONTEXT

Higher HSMR regions may be associated with:

- Strained hospital capacity
- Workforce shortages
- Delays in access to care



RECOMMENDATIONS

- Target high-HSMR provinces (>110): NL, PE, MB
- Strengthen real-time monitoring systems
- Address workforce and capacity gaps
- Standardize clinical practices across provinces



IMPLICATIONS FOR DECISION-MAKERS

For provincial health authorities:

- Prioritize intervention in high-risk regions
- Allocate resources based on data-driven insights
- Implement targeted quality improvement initiatives



CONCLUSION

Although hospital mortality trends appear stable, significant regional disparities persist. Targeted, data-driven interventions are essential to reduce inter-provincial differences and improve patient outcomes across Canada.



Better Data. Better Decisions. Better Outcomes.

Source: CIHI Hospital Standardized Mortality Ratio (HSMR) Reports (2020–2025).